

Civic-Connect: Project Engineering Document (PED) v1.0

Milestone 1: Engineering Foundation & Requirements Baseline

ENGINEERING LARPERS

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Documentation:

Version	Date	Purpose	Author/Reviewer
1.0	9 Sep 2026	Initial M1 baseline: problem, stakeholders, scope, requirements, constraints, risk, governance, sign-off	Engineering Larpers

Problem and Business Need

Problem Statement

CivicConnect currently manages community service requests through a fragmented mix of channels — email, telephone, WhatsApp messages, spreadsheets and paper records. Because no single controlled record exists, requests are frequently duplicated, overlooked, incorrectly assigned or lost between channels. Requesters have little visibility into whether a request has been received, assigned, delayed, resolved or closed, which erodes trust in the organisation's service delivery. Staff struggle to prioritise work, establish clear ownership of individual requests, and coordinate effort across the team, while accountability for status changes and actions taken is weak because there is no single traceable history per request. Management, in turn, lacks reliable, consolidated information on outstanding, overdue and resolved work, making reporting manual, inconsistent and difficult to audit.

Business Need

CivicConnect needs a controlled digital platform that gives requesters, staff and management a single, reliable and traceable way to submit, manage, monitor and report on service requests — improving visibility and accountability across the request lifecycle without introducing an unsustainable technical, operational or financial burden on the organisation.

Stakeholder Analysis

Three primary stakeholder groups were identified from the CivicConnect scenario. Their needs, and their relative influence and interest in the platform, are summarised below.

Stakeholder	Main Needs	Influence / Interest
Requesters	Submit service requests; track status; view request history; receive meaningful, timely	Medium influence / High interest

	feedback when a request is accepted, updated or completed.	
Staff	View and search relevant requests; assign or accept responsibility; update status through controlled transitions; record actions/comments; resolve or close requests efficiently.	High influence / High interest
Management	Monitor outstanding, overdue, resolved and closed requests; view activity by category/status; access enough information to support accountability and service-performance reporting.	High influence / High interest

Stakeholder Conflict / Competing Expectations

The clearest tension identified is between requester transparency and staff workload control. Requesters want frequent, granular visibility into their request (who it is assigned to, why it is delayed, what is being done), because this directly drives their trust in the service. Staff, however, need the freedom to triage, batch and prioritise work without being interrupted by constant status queries, and are wary of a system that exposes internal workload or assignment decisions in a way that invites pressure from requesters or management.

This conflict was resolved for the M1 baseline by scoping requester-facing status visibility to a small set of controlled, staff-driven status values (e.g. Submitted → Assigned → In Progress → Resolved → Closed) rather than free-text or real-time internal detail. This satisfies the requester's core need for visibility (FR-002) while preserving staff control over how and when status changes are communicated (FR-005) and is reflected as a deliberate scope decision the team can defend at the M1 baseline sign-off.

Scope Baseline

In-Scope

All FRs identified are committed to the M1 baseline:

ID	Capability	FR Link	Priority
S-001	Request Submission	FR-001	High
S-002	Status Tracking	FR-002	High
S-003	Staff Request Queue	FR-003	High

S-004	Request Assignment	FR-004	High
S-005	Status Transitions	FR-005	High
S-006	Action/Comments Logging	FR-006	Medium
S-007	Management Monitoring	FR-007	Medium

Out-of-Scope

ID	Exclusion	Rationale	Stakeholder Impact
OS-001	Mobile Application (Native)	FRs assume web-based interaction, mobile would require separate codebases (iOS/Android), testing overhead, and ongoing operational costs. The team lacks expertise in mobile development.	Acceptable - responsive web application meets 80% of user needs for M1. Management approved web-only for Phase 1.
OS-002	Email/WhatsApp Integration	Not covered in FRs. Would require complex API integration, message parsing, and authentication challenges with significant security and reliability risks.	Acceptable - requesters will be onboarded to the web platform.
OS-003	Real-time Notifications (SMS/Push)	FRs do not require push notifications. Real-time notifications require third-party SMS gateways or push services with ongoing operational costs not approved in the budget.	Acceptable - feedback provided via web interface and email updates only.
OS-004	Advanced Analytics / BI Reports	FR-007 only requires basic operational metrics. Complex reporting and data visualisation tools are outside current scope.	Acceptable - management accepts basic dashboard for M1, with advanced reporting deferred.
OS-005	Multi-language Support	Not identified as a stakeholder need in stakeholder analysis. Multi-language support would add significant UI, validation, and error messaging complexity without immediate stakeholder value.	Acceptable - staff confirmed English-only is acceptable for initial deployment.

Deferred / Future Scope

ID	Deferred Item	Reason for Deferral
DS-001	Mobile Application	Requires separate platform development and user research.
DS-002	Email Integration (Inbound)	Allows time to evaluate email parsing libraries and security implications.
DS-003	SLA (Service Level Agreement) Reporting	Requires operational data to define meaningful SLAs.
DS-004	Automated Escalation Rules	Depends on actual usage patterns and staff workflows observed post-deployment.
DS-005	Bulk Request Import (CSV/Excel)	Staff capability to batch-create requests for maintenance events.

Requirements and Acceptance Criteria

Functional Requirement

The functional requirements below are derived directly from the stakeholder needs identified in Section 2. Each requirement has a unique identifier, an explicit source stakeholder, an indicative priority, and a testable acceptance criterion, in line with the Master Project Brief's requirements standard (Section 11).

ID	Functional Requirement	Source	Priority	Acceptance Criterion
FR-001	The system shall allow requesters to submit a new service request with the required information.	Requester	High	A requester can submit a request and the system creates a unique request record.
FR-002	The system shall allow requesters to view the current status of their submitted requests.	Requester	High	A requester can select a submitted request and see its current, controlled status.
FR-003	The system shall allow authorised staff to view service requests relevant to them.	Staff	High	An authorised staff member can access their relevant requests after logging in.
FR-004	The system shall allow authorised staff to assign or accept responsibility for a request.	Staff	High	A staff member can assign/accept a request, and the responsible person is recorded.
FR-005	The system shall allow authorised staff to update the status of a request through controlled status transitions.	Staff	High	A permitted status change updates the request and records the new status and timestamp.

FR-006	The system shall allow staff to record comments, actions or resolution information against a request.	Staff	Medium	A staff member can add information to a request, and it is saved against that request, retaining its author and time.
FR-007	The system shall allow management to identify open, overdue, resolved and closed requests.	Management	Medium	Management can view requests filtered or grouped by these statuses.

Priority reflects the requirement's contribution to the core problem (visibility, accountability and traceability) and is a draft assignment for team confirmation ahead of baseline sign-off; it is not yet linked to a formal MoSCoW or numeric scoring exercise.

Non-Functional Requirement

Overview

All NFRs are derived from stakeholder needs and the project's constraints identified in the Constraints Analysis document. Each NFR is **measurable** - written with specific targets and includes acceptance criteria.

NFR-001: Response Time (Performance)

Attribute	Detail
ID	NFR-001
Description	The requester dashboard must load within 2 seconds when accessed by a user on a standard broadband connection.
Source	Requester needs: "Submit requests, track status, view history" requesters need a responsive system.
Priority	High
Measurement	Use Chrome DevTools Network tab to measure Time to First Byte (TTFB) and full-page load time. Test with 3 representative pages: Dashboard, Request List, and Request Detail. Run 10 consecutive measurements.
Acceptance Criteria	AC-NFR-001: Dashboard page loads in < 2 seconds in 95% of test runs (10 consecutive measurements).
How It Constrains M2 Architecture	Forces the team to consider caching database indexing, efficient API design (avoid N+1 queries), and server-side rendering vs. client-side rendering trade-offs. Technology choice must support these patterns.

Downstream Risk	If the selected frontend framework is heavy, or the database is not indexed, performance will fail. Requires performance testing in M3. If free-tier database performance is insufficient, may need to increase threshold to 3 seconds or implement more aggressive caching.
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NFR-002: Concurrent Users (Scalability)

Attribute	Detail
ID	NFR-002
Description	The system must support at least 50 concurrent users without significant degradation. "Significant degradation" defined as response time > 4 seconds on the dashboard.
Source	Staff and Management needs stakeholder analysis: "Multiple staff working simultaneously" - the system must handle real-world usage.
Priority	Medium
Measurement	Use Apache JMeter or k6 to simulate 50 concurrent users performing typical actions (view dashboard, submit request, update status) over a 5-minute period.
Acceptance Criteria	AC-NFR-002: All requests complete successfully (HTTP 200) with average response time < 3 seconds and maximum < 5 seconds.
How It Constrains M2 Architecture	Forces the team to consider database connection pooling, stateless application design for horizontal scaling, and load balancing. Technology must support multiple concurrent connections (e.g., Node.js event loop, or Java thread pooling).
Downstream Risk	If the team chooses a framework with poor concurrency support (e.g., PHP without Apache tuning), or a database with low connection limits (e.g., free-tier PostgreSQL with 20 connections), this NFR will fail. Free-tier limitations are a key concern (see Trade-Off 2 in Constraints Analysis).

NFR-003: Data Encryption (Security)

Attribute	Detail
ID	NFR-003
Description	All data transmitted between the client and server must be encrypted using TLS 1.2 or higher.
Source	Security constraint from Master Project Brief (SE-001): "Security is a lifecycle-wide responsibility" and SE-003: "Data in transit must be encrypted."
Priority	High

Measurement	Use SSL Labs SSL Server Test (or equivalent tool) to verify the deployed site uses TLS 1.2+ and has no known vulnerabilities (e.g., POODLE, Heartbleed).
Acceptance Criteria	AC-NFR-003: SSL Labs test returns an overall rating of "A" or "A+".
How It Constrains M2 Architecture	Forces the team to choose a deployment platform that provides free SSL certificates. Cannot use plain HTTP. Must enforce HTTPS redirection at the application or server level.
Downstream Risk	If the selected platform doesn't support automatic SSL (e.g., some free hosting services), the team must manually configure certificates, creating an operational burden.

NFR-004: Audit Logging (Accountability)

Attribute	Detail
ID	NFR-004
Description	All status changes on service requests must be logged with: timestamp, user ID, previous status, new status, and optional comment.
Source	Staff and Management needs stakeholder analysis. "Accountability" and the Master Project Brief's problem statement about "weak accountability for changes to request status."
Priority	High
Measurement	Manually verify audit log entries in the database after performing status transitions. Query the audit table to check completeness of fields.
Acceptance Criteria	AC-NFR-004: After 5 distinct status transitions, the audit log contains 5 entries with all required fields populated (timestamp, user ID, previous status, new status).
How It Constrains M2 Architecture	Forces the team to design the database schema to support an audit log table. The architecture must ensure that every status update triggers an audit record - this may be implemented via database triggers, ORM hooks, or explicit service-layer logic. Technology choice must support database transactions to ensure atomicity (status update and audit log must succeed or fail together).
Downstream Risk	If the team forgets to implement audit logging early, retrofitting it later will be difficult and error prone. Must be designed in M2 and implemented in M3.

NFR-005: Availability (Reliability)

Attribute	Detail
ID	NFR-005
Description	The system must be available 99.5% of the time during business hours.
Source	Staff and Management needs stakeholder analysis. "Monitor and analyse performance" - you can't monitor or analyse if the system is down.
Priority	Medium

Measurement	Monitor uptime using UptimeRobot or similar tool over a 2-week period during business hours.
Acceptance Criteria	AC-NFR-005: Uptime $\geq 99.5\%$ (maximum 43 minutes of downtime per month during business hours) over a 2-week test period.
How It Constrains M2 Architecture	Forces the team to consider deployment platform reliability (choose a reputable cloud provider with good SLAs), error handling (graceful failure), database backup/recovery, and health check endpoints for monitoring. Technology should support hot-reload or zero-downtime deployment if possible.
Downstream Risk	Free-tier platforms may have less reliability (e.g., Heroku dyno sleeps after 30 minutes idle). The team may need to choose a platform with better uptime guarantees, potentially incurring cost (see Trade-Off 2 in Constraints Analysis).

NFR-006: Accessibility (Usability)

Attribute	Detail
ID	NFR-006
Description	The web interface must meet basic WCAG 2.1 Level A accessibility standards.
Source	Organisation's inclusivity policy (assumed/implicit quality expectation) and good engineering practice.
Priority	Low
Measurement	Use automated accessibility checker (e.g., axe DevTools, Lighthouse) to scan all major pages: Dashboard, Request Form, Request List, and Request Detail pages.
Acceptance Criteria	AC-NFR-006: No Level A violations on Dashboard, Request Form, Request List, and Request Detail pages as reported by axe DevTools.
How It Constrains M2 Architecture	Forces the team to choose a frontend framework with good accessibility support (e.g., semantic HTML, ARIA attributes). Design decisions must consider colour contrast, keyboard navigation, and screen reader support.
Downstream Risk	If the team uses a UI library that doesn't prioritise accessibility (e.g., some custom component libraries), they'll need to manually add ARIA roles, increasing development time.

Constraints

Scope Constraints

ID	Constraint	Description	Engineering Implication
SC-001	Fixed Scope	All 7 FRs (FR-001 to FR-007) identified are committed to the baseline.	Cannot drop any FRs; any scope change requires formal change control with impact analysis.
SC-002	Minimum Capabilities	The system must support all Minimum Business Capabilities defined in the Master Project Brief.	Cannot drop core features; prioritisation applies to additional features only.

SC-003	No Unapproved Additions	Additional features are permitted only if justified by stakeholder value and effect on scope/schedule/quality/security/risk is considered.	Each proposed new FR requires impact analysis; "nice-to-have" features are deferred to DS-001 to DS-005.
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Schedule Constraints

ID	Constraint	Description	Engineering Implication
SH-001	4 Milestones	Four formal milestones within the SEN381 delivery period: M1 (Week 4), M2 (Week 8), M3 (Week 12), M4 (Week 16).	Cannot delay M1 without affecting downstream phases. All 7 FRs and 6 NFRs must be deliverable within the fixed timeline.
SH-002	Fixed End Date	No extension beyond academic calendar.	Scope must be realistic; defer non-essential features (DS-001 to DS-005) to maintain quality.
SH-003	Review Lead Time	Two-reviewer approval required for substantive changes entering main.	Team must plan for review cycles; cannot rush changes at the last minute.

Cost / Resource Constraints

ID	Constraint	Description	Engineering Implication
CR-001	Free/Low-Cost Services	Prefer free or low-cost services where practical; identify limitations.	Technology choices must evaluate free-tier limitations like deployment platform free tier, database size limits, connection limits. Must support all 7 FRs and 6 NFRs on free tier.
CR-002	Team Size	Exactly 3 students; no additional paid developers.	Work must be divided efficiently; complexity must match team capacity.
CR-003	Operational Cost	Likely operational cost beyond educational context must be identified.	Hosting, domain, third-party API costs must be documented and reasonable.
CR-004	Learning Curve	Unfamiliar technologies introduce schedule risk.	Technology selection must consider team's existing skills; if adopting new tech, factor in learning time.

Quality Constraints

ID	Constraint	Description	Engineering Implication
QU-001	Measurable Quality	Quality attributes must be defined and supported by measurable evidence	All NFRs (NFR-001 to NFR-006) are written with measurable targets.
QU-002	Acceptance Criteria	Important requirements must have acceptance criteria.	Every FR and every NFR has defined acceptance criteria.
QU-003	Test Evidence	Test coverage alone is insufficient; evidence must show what was verified and what was not.	Unit tests, integration tests, and manual tests must be traceable to requirements in the RTM.
QU-004	Defect Management	Defects must be recorded with severity/priority and disposition.	Cannot ignore bugs; must track and address them.

Security Constraints

ID	Constraint	Description	Engineering Implication
SE-001	Lifecycle Responsibility	Security is a lifecycle-wide responsibility, not a final add-on.	NFR-003 (encryption) and NFR-004 (audit logging) defined in M1; security must be considered in M2 architecture, M3 implementation, and M4 deployment.
SE-002	Secrets Handling	Passwords, API keys, tokens, private keys must not be committed to GitHub.	Must use environment variables and .gitignore; enforce with GitHub pre-commit hooks.
SE-003	Data in Transit	All sensitive data (requests, user info) must be encrypted in transit (TLS 1.2+).	Technology choice must support HTTPS; deployment platform must provide SSL certificates (NFR-003).
SE-004	Authentication & RBAC	Authentication and Role-Based Access Control are mandatory.	Cannot skip authentication for simplicity; must design RBAC from the start (supports FR-003, FR-004, FR-007).
SE-005	Residual Risks	Cannot claim system is "secure"; record residual security risks.	Threat modelling must identify weaknesses; risk register must document accepted risks.

Initial Requirements Traceability Matrix

NFR Traceability

NFR ID	Source	Acceptance Criteria	Design Link (M2)	Test Link (M3)
NFR-001	Requester	AC-NFR-001: Dashboard loads in < 2s (95% of tests)	TBD	TBD
NFR-002	Staff/Management	AC-NFR-002: 50 concurrent users, avg < 3s, max < 5s	TBD	TBD
NFR-003	Security Constraint	AC-NFR-003: SSL Labs rating A/A+	TBD	TBD
NFR-004	Staff/Management	AC-NFR-004: All status changes logged with required fields	TBD	TBD
NFR-005	Staff/Management	AC-NFR-005: 99.5% uptime during business hours	TBD	TBD
NFR-006	Org Policy (implicit)	AC-NFR-006: No WCAG 2.1 Level A violations	TBD	TBD

Constraint Traceability

Constraint ID	Source	Engineering Implication	M2 Impact	M3 Impact
SC-001	Master Project Brief	Fixed scope; change control required	Design decisions must be reversible/adaptable	Changes require impact analysis
SC-002	Master Project Brief	Minimum capabilities must be delivered	Architecture must support all FRs	All FRs must be implemented
SC-003	Master Project Brief	No unapproved additions	Each proposed feature requires justification	No unapproved scope creep
SH-001	Master Project Brief	4 milestones, fixed dates	Must deliver architecture by M2	Must deliver working code by M3
SH-002	Master Project Brief	Fixed end date	Realistic scope; defer non-essential features	Must complete by deadline
SH-003	Master Project Brief	2 approvals for PRs	Plan for review cycles	Cannot rush changes
CR-001	Master Project Brief	Free/low-cost services	Platform selection constrained by free-tier limits	Deployment platform must be free/compatible
CR-002	Master Project Brief	Team of 3	Complexity must match team capacity	Work must be divided efficiently
CR-003	Master Project Brief	Operational cost	Document projected costs	Include cost review in final
CR-004	Team Analysis	Learning curve	Factor learning time into planning	Monitor learning curve impact

QU-001	Master Project Brief	NFRs must be measurable	Architecture must support performance testing	Tests must validate NFRs
QU-002	Master Project Brief	Acceptance criteria	Design must meet ACs	Tests must verify ACs
QU-003	Master Project Brief	Test evidence	Architecture must support testability	Tests must be traceable to requirements
QU-004	Master Project Brief	Defect management	Track defects	Address defects systematically
SE-001	Master Project Brief	Security lifecycle-wide	Security patterns built into architecture	Security scanning in CI/CD
SE-002	Master Project Brief	Secrets handling	Environment variables in architecture	No secrets in GitHub
SE-003	Master Project Brief	Data in transit encryption	Platform must support HTTPS	TLS enforced
SE-004	Master Project Brief	Authentication & RBAC	Design RBAC from the start	Implement authentication
SE-005	Master Project Brief	Residual risks	Document accepted risks	Record residual security risks

Initial Risk Register

ID	Description	Cause	Prob.	Impact	Priority	Mitigation	Contingency	Owner	Status
R-001	Team lacks prior experience with the backend framework likely to be chosen in M2	Skills gap / learning curve	Medium	High	High	Timebox a technical spike in the first week of M2 before committing	Fall back to a framework the team already knows from prior modules	Busiswe & Tshegofatso	Open
R-002	Stakeholder (management) requests features beyond the agreed MVP scope	Enthusiastic but uncontrolled stakeholder input	Medium	Medium	Medium	Apply the Change Request/Impact Analysis template (Master Brief Appendix E) to any new ask	Log as deferred/future scope if it cannot be absorbed without schedule impact	Katlego	Open
R-003	Free-tier hosting/database limits are exceeded during load testing or demo	Under-provisioned free-tier resources	Low	Medium	Medium	Monitor usage against documented free-tier limits from M2 onward	Migrate to a low-cost paid tier or reduce load-test scale for the demo	Busiswe & Tshegofatso	Open
R-004	Two-reviewer approval is skipped under deadline pressure close to a milestone	Time pressure vs. governance discipline	Medium	Medium	Medium	Enforce branch protection technically in GitHub settings, not	Retrospective review of merge history before each milestone submission	Busiswe & Tshegofatso	Open

						just by agreement			
R-005	A baselined requirement is edited after sign-off without a formal change record	Informal communication overriding controlled process	Low	High	Medium	Team agreement: all baseline edits go through the Change Request template	Revert to last baselined version and reprocess the change formally	Requirements Engineer	Open

Forward Engineering Consideration

Concern	Why it matters now	Later decision/activity it influences	Evidence still missing	Risk of ignoring it
Security (RBAC & data protection)	Shapes the data model and authentication requirements before they're designed.	Authentication mechanism/technology choice in M2.	Which specific permissions each of the 3 roles requires, in detail.	Bolted-on security in M3 causes rework and weakens the security-evidence marks.
Testability	NFRs must already be written so they can be tested later, not rewritten in M3.	Test framework/tooling choice, aligned to the M2 technology decision.	Which stack the team will choose (deferred, see Decision D-002).	Untestable/vague requirements block quality evidence at M3.
Deployment environment	Availability (NFR-004 candidate) and environment parity depend on where the system will run.	Hosting platform decision in M2.	Team's budget ceiling and familiarity with candidate platforms.	Late discovery of platform incompatibility disrupts the M3 release schedule.
Data backup/recovery	Even an MVP holds real (if simulated) service-request data that requesters rely on.	Persistence and backup design in M2/M3.	A concrete data-retention/recovery policy.	Data loss with no recovery path undermines the whole platform's credibility.
Observability	Management's dashboards need (FR-006) implies logging/monitoring must exist, not just a UI.	Logging/monitoring tool selection.	Which metrics matter most to management oversight.	No way to demonstrate system health or diagnose issues at M4.
Scalability	Expected concurrent user load is still unknown and affects NFR-001.	Architecture pattern choice in M2.	A realistic estimate of concurrent requesters/staff.	Performance NFR silently fails under real usage, discovered too late.
Maintainability	Three people with individually assessed accountability must all be able to read each other's code.	Coding standard/architecture pattern agreed in M2.	An agreed style guide/lint configuration.	Inconsistent code increases technical debt and slows M3 review.

Engineering Decision Log

ID	Context	Constraints	Alternatives	Decision	Rationale	Trade-offs	Risks	Evidence	Later consequence
D-001	Need version-control governance in place before any controlled artefact work begins.	Master Project Brief mandates a protected main branch and a minimum of two reviewer approvals.	(a) Single shared main branch, no protection, fast but ungoverned. (b) Protected main with mandatory 2-reviewer PRs.	Option (b) adopted.	Required by the Master Brief; enforces a real quality gate rather than a documentation exercise.	Slower merge velocity with only 3 available reviewers (author can never self-approve).	Risk of review becoming a bottleneck or a rubber stamp under deadline pressure (see R-004).	GitHub branch-protection settings screenshot + PR history.	Revisit at M3 once real PR volume is known.
D-002 (deferred)	Team must eventually choose a backend/frontend technology stack.	M1 explicitly excludes final technology-stack selection as a required decision.	Not yet meaningfully explored — no quality-attribute prioritisation or skills inventory has been done yet.	DEFERRED to Milestone 2.	Insufficient evidence currently exists to justify a defensible choice; deciding now would be a guess, not an engineering decision.	—	Risk R-001 (skills gap) is directly tied to whichever stack is eventually chosen.	—	To be decided once architecturally significant requirements and quality attributes are identified in M2.

GitHub and Team Governance*

1. One controlled team repository, structured per the Master Brief's suggested layout (docs/PED, requirements, risk, decisions, architecture, change; src; tests; GitHub/workflows).
2. Main branch protected: direct pushes disabled; all substantive changes to controlled artefacts or code enter via Pull Request.
3. Minimum two approvals required per PR, from reviewers other than the author; self-approval disabled in repository settings.
4. Issues/board used to track engineering work (documentation and registers, not only application code) from the start of M1, so repository history shows authentic progressive participation from all three members.
5. No secrets, tokens or credentials committed to the repository; a .gitignore and environment-variable convention were agreed before any configuration work began.
6. [GitHub Repo Access](#)

AI Usage Register*

Date	Student	Tool	Engineering task	AI contribution	Verification	Decision	Issues found
3 Sep 2026	Katlego	Claude	Drafting initial FR-001–FR-007 and NFR006	Suggested first-pass phrasing and possible acceptance-criteria structure	Cross-checked each suggestion against the stakeholder analysis; rewrote vague phrasing to be testable	Accepted with edits	Initial AI draft used non-measurable language (“system should be fast”); replaced with numeric criteria

4 Sep 2026	Busisiwe	Claude	Drafting Stakeholder Analysis and Scope	Suggested generic stakeholder responses and in reach scope categories	Replaced generic entries with CivicConnect-specific scope tied to this team's actual constraints	Accepted with edits	Generic AI-suggested scope were not specific enough on their own; rewritten with causes/owners
5 Sep 2026	Tshegofatso	Claude	Drafting the initial Risk Register and Forward Engineering Consideration	Suggested generic risk categories (schedule, technical, security, resourcing)	Replaced generic entries with CivicConnect-specific risks tied to this team's actual constraints	Accepted with edits	Generic AI-suggested risks were not specific enough on their own; rewritten with causes/owners
6 Sep 2026	Tshegofatso	Claude	Drafting PED document formatting	Suggested easy format derived from M1 document and Project document	Crossed-checked table formatting in M1 document and fact checked through different free-online PEDs	Accepted with edits	Initial AI draft used non-measurable language ("system should be fast"); replaced with numeric criteria

PED v1.0 Baseline Sign-Off*

Baseline Type	Milestone 1 — Engineering Foundation & Requirements Baseline (PED v1.0)
Version	v1.0
Date	9 Sep 2026
Scope reviewed	YES
Requirements/traceability checked	YES
Risk review completed	YES
Repository/governance controls checked	YES
Outcome	ACCEPTED (example outcome — your team's assessor records the real decision)